EXPERIMENT 10

Subqueries and Correlated Subqueries

Aim: To retrieve data using a subquery and a correlated subquery.

Software Required: MySQL 8.0 Command Line Client

SQL Program

```
USE SQL_Lab;

CREATE TABLE Employee_Salary (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    department VARCHAR(20),
    salary INT
);

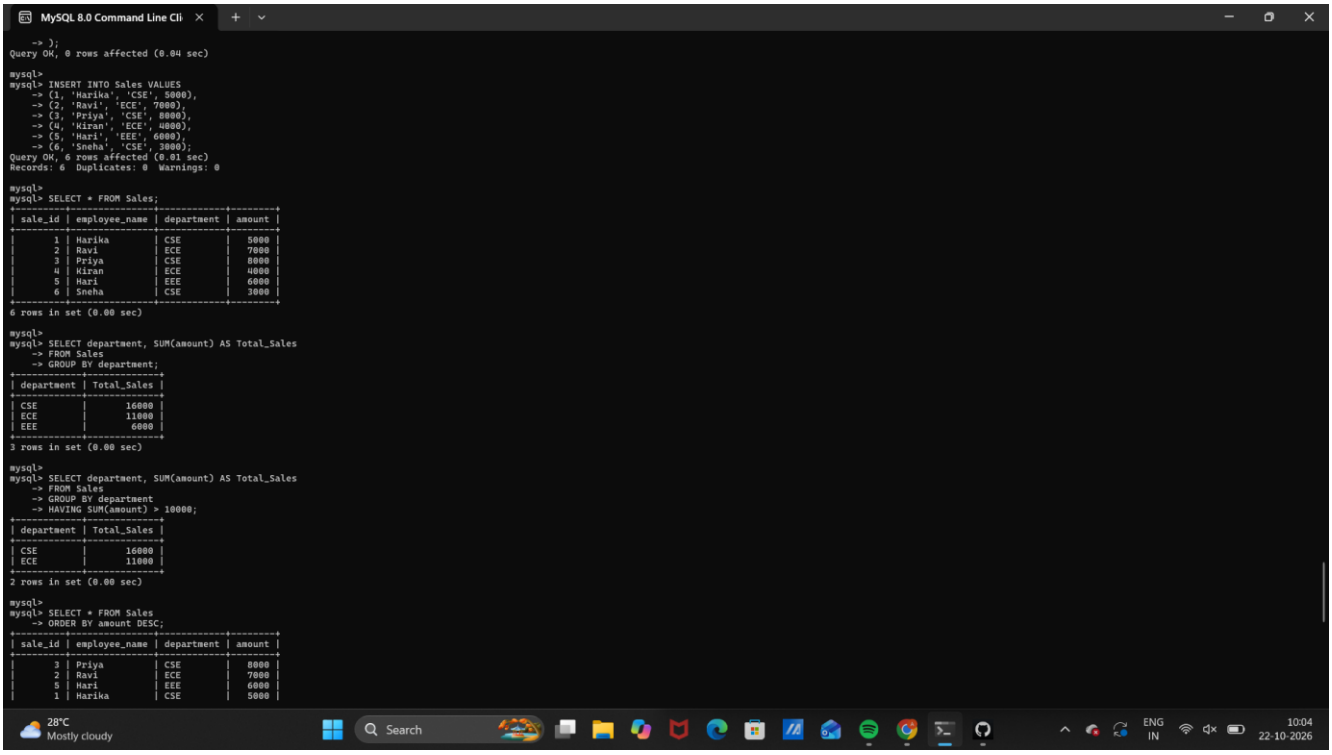
INSERT INTO Employee_Salary VALUES
(1, 'Harika', 'CSE', 40000),
(2, 'Ravi', 'ECE', 35000),
(3, 'Priya', 'CSE', 50000),
(4, 'Kiran', 'ECE', 30000),
(5, 'Hari', 'EEE', 45000);

SELECT * FROM Employee_Salary;

SELECT emp_name, salary
FROM Employee_Salary
WHERE salary > (
    SELECT AVG(salary)
    FROM Employee_Salary
);

SELECT e.emp_name, e.department, e.salary
FROM Employee_Salary e
WHERE e.salary > (
    SELECT AVG(e2.salary)
    FROM Employee_Salary e2
    WHERE e2.department = e.department
);
```

Expected Output



```
MySQL 8.0 Command Line Cli

    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> INSERT INTO Sales VALUES
    -> (1, 'Harika', 'CSE', 5000),
    -> (2, 'Ravi', 'ECE', 7000),
    -> (3, 'Priya', 'CSE', 8000),
    -> (4, 'Kiran', 'ECE', 4000),
    -> (5, 'Hari', 'EEE', 6000),
    -> (6, 'Sneha', 'CSE', 3000);
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

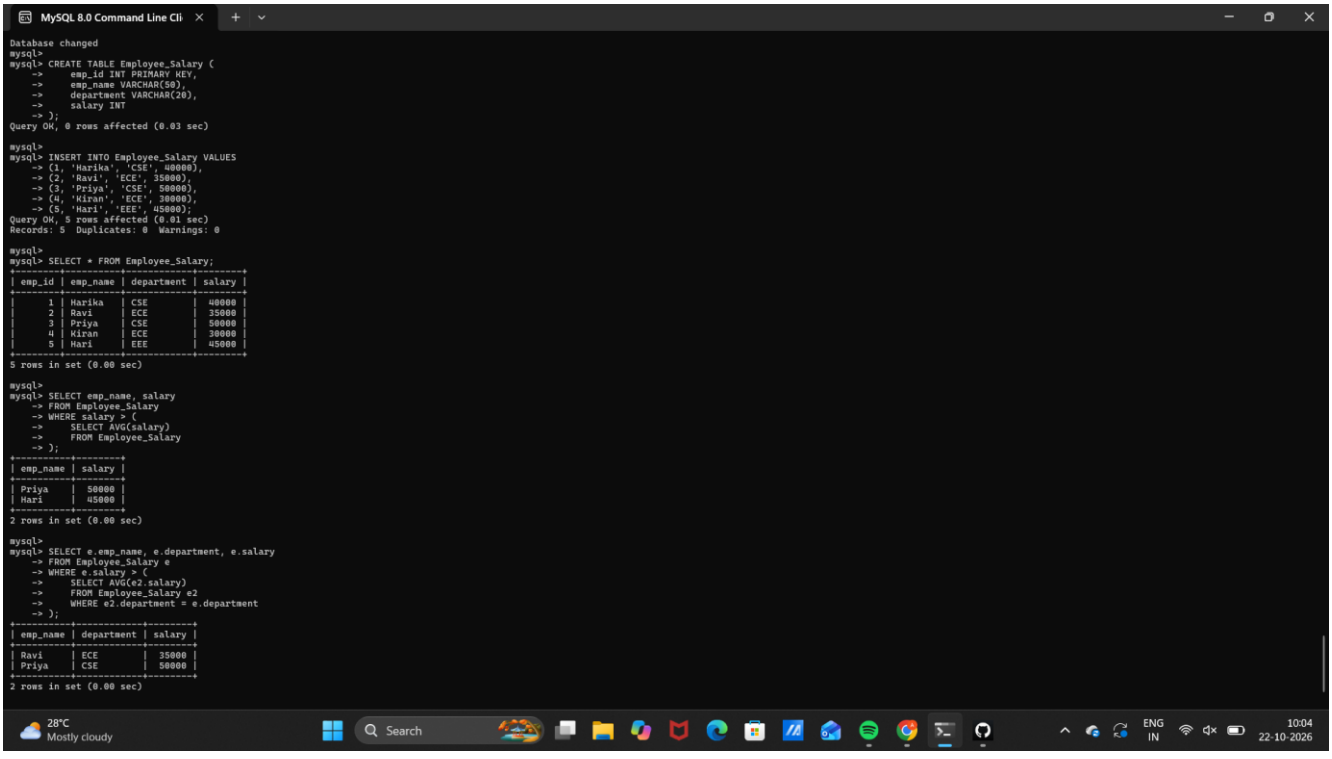
mysql>
mysql> SELECT * FROM Sales;
+---------+---------------+------------+--------+
| sale_id | employee_name | department | amount |
+---------+---------------+------------+--------+
|       1 | Harika        | CSE        |   5000 |
|       2 | Ravi          | ECE        |   7000 |
|       3 | Priya         | CSE        |   8000 |
|       4 | Kiran         | ECE        |   4000 |
|       5 | Hari          | EEE        |   6000 |
|       6 | Sneha         | CSE        |   3000 |
+---------+---------------+------------+--------+
6 rows in set (0.00 sec)

mysql>
mysql> SELECT department, SUM(amount) AS Total_Sales
    -> FROM Sales
    -> GROUP BY department;
+------------+-------------+
| department | Total_Sales |
+------------+-------------+
| CSE        |       16000 |
| ECE        |       11000 |
| EEE        |        6000 |
+------------+-------------+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT department, SUM(amount) AS Total_Sales
    -> FROM Sales
    -> GROUP BY department
    -> HAVING SUM(amount) > 10000;
+------------+-------------+
| department | Total_Sales |
+------------+-------------+
| CSE        |       16000 |
| ECE        |       11000 |
+------------+-------------+
2 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM Sales
    -> ORDER BY amount DESC;
+---------+---------------+------------+--------+
| sale_id | employee_name | department | amount |
+---------+---------------+------------+--------+
|       3 | Priya         | CSE        |   8000 |
|       2 | Ravi          | ECE        |   7000 |
|       5 | Hari          | EEE        |   6000 |
|       1 | Harika        | CSE        |   5000 |
+---------+---------------+------------+--------+

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Database changed
mysql>
mysql> CREATE TABLE Employee_Salary (
    ->     emp_id INT PRIMARY KEY,
    ->     emp_name VARCHAR(50),
    ->     department VARCHAR(20),
    ->     salary INT
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> INSERT INTO Employee_Salary VALUES
    -> (1, 'Harika', 'CSE', 40000),
    -> (2, 'Ravi', 'ECE', 35000),
    -> (3, 'Priya', 'CSE', 50000),
    -> (4, 'Kiran', 'ECE', 30000),
    -> (5, 'Hari', 'EEE', 45000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT * FROM Employee_Salary;
+--------+----------+------------+--------+
| emp_id | emp_name | department | salary |
+--------+----------+------------+--------+
|      1 | Harika   | CSE        |  40000 |
|      2 | Ravi     | ECE        |  35000 |
|      3 | Priya    | CSE        |  50000 |
|      4 | Kiran    | ECE        |  30000 |
|      5 | Hari     | EEE        |  45000 |
+--------+----------+------------+--------+
5 rows in set (0.00 sec)

mysql>
mysql> SELECT emp_name, salary
    -> FROM Employee_Salary
    -> WHERE salary > (
    ->     SELECT AVG(salary)
    ->     FROM Employee_Salary
    -> );
+----------+--------+
| emp_name | salary |
+----------+--------+
| Priya    |  50000 |
| Hari     |  45000 |
+----------+--------+
2 rows in set (0.00 sec)

mysql>
mysql> SELECT e.emp_name, e.department, e.salary
    -> FROM Employee_Salary e
    -> WHERE e.salary > (
    ->     SELECT AVG(e2.salary)
    ->     FROM Employee_Salary e2
    ->     WHERE e2.department = e.department
    -> );
+----------+------------+--------+
| emp_name | department | salary |
+----------+------------+--------+
| Ravi     | ECE        |  35000 |
| Priya    | CSE        |  50000 |
+----------+------------+--------+
2 rows in set (0.00 sec)

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Observation

Execute the SQL statements in MySQL 8.0 Command Line Client. Verify that each statement runs successfully and that the displayed rows match the inserted sample data and query conditions.

Note: The first query returns employees earning more than the overall average salary. The correlated query compares each employee with their own department average.

Result

Thus, Experiment 10 (Subqueries and Correlated Subqueries) was performed using MySQL 8.0.